

CA-C30L: Mobile Application Development Lab

LAB PROGRAM-10 : SENDING SMS

2. Create an application to send SMS and receive SMS

Note : This program will run in emulator or LD player but sms can be only sent to phone after converting to apk and installing in phone

Program explanation:

This program uses

1. Androidmanifest.xml -> Permissions

- android.permission.SEND_SMS: Needed to send SMS messages.
- <uses-feature android.hardware.telephony>: Declares that the app uses telephony features.

2. Activitymain.xml->following views for design

Two EditText views:

- editText1: For entering the recipient's phone number.
- editText2: For typing the SMS message.

Button:

- btnSendSMS: Sends the message when clicked.

3. MainActivity.java->following

- **Permissions:** Requests SEND_SMS permission at runtime.

```
ActivityCompat.requestPermissions(this, new  
String[] {Manifest.permission.SEND_SMS},  
PackageManager.PERMISSION_GRANTED);
```

- **UI References:**

```
et1 = findViewById(R.id.editText1); // phone number  
et2 = findViewById(R.id.editText2); // message  
btn1 = findViewById(R.id.btnSendSMS); // send button
```

- **Button Click Listener:**

When the button is clicked:

```
String number = et1.getText().toString();
String sms = et2.getText().toString();
SmsManager smsmanager = SmsManager.getDefault();
smsmanager.sendTextMessage(number, null, sms, null, null);
```

- Gets the phone number and message.
- Sends the SMS using SmsManager.
- **Toast Message:** Shows "Message sent success" on successful sending.

(Write the following highlighted code in android manifest file and activity_main xml files and entire code of java file)

Android Manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest
xmlns:android="http://schemas.android.com/apk/res/
android" package="com.example.smsdemo2">

    <uses-feature
        android:name="android.hardware.telephony"
        android:required="false" />
    <uses-permission
        android:name="android.permission.SEND_SMS" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Base.Theme.SMSDemo2">

        <activity
            android:name="com.example.smsdemo2.MainActivity"
            android:exported="true">
            <intent-filter>
                <action
                    android:name="android.intent.action.MAIN" />
```

```
        <category
android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>
</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/
android"
xmlns:app="http://schemas.android.com/apk/res-
auto"
xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
<EditText
    android:id="@+id/editText1"
    android:layout_width="wrap_content"
    android:layout_height="100dp"
    android:layout_marginTop="56dp"
    android:ems="10"
    android:hint="Enter your Phone number"
    android:inputType="text"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.477"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toTopOf="parent" />

    <EditText
    android:id="@+id/editText2"
    android:layout_width="240dp"
```

```
        android:layout_height="247dp"
        android:ems="10"
        android:hint="Enter your message"
        android:inputType="text"

app:layout_constraintBottom_toBottomOf="parent"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.497"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/editText1"

app:layout_constraintVertical_bias="0.17" />

        <Button
            android:id="@+id/btnSendSMS"
            android:layout_width="wrap content"
            android:layout_height="wrap content"
            android:layout_marginBottom="200dp"
            android:text="Send sms"

app:layout_constraintBottom_toBottomOf="parent"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.32"

app:layout_constraintStart_toStartOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.smsdemo2;

import android.Manifest;
import android.annotation.SuppressLint;
```

```
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

public class MainActivity extends
AppCompatActivity {

    EditText et2;
    EditText et1;
    Button btn1;
    @SuppressWarnings("MissingInflatedId")
    @Override
protected void onCreate(Bundle
savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    //request permission for sms
    ActivityCompat.requestPermissions(this, new
String[] {Manifest.permission.SEND_SMS},
PackageManager.PERMISSION_GRANTED);
    //create objects of views
    btn1 = (Button)
findViewById(R.id.btnSendSMS);
    et1 =
(EditText) findViewById(R.id.editText1);
    et2 = (EditText)
findViewById(R.id.editText2);

    //on click listener
    btn1.setOnClickListener(new
View.OnClickListener() {
        @Override
        public void onClick(View v) {
```

```
        String number =
et1.getText().toString();//fetching values of
phone no. and message
        String
sms=et2.getText().toString();
        try{
            SmsManager smsmanager =
SmsManager.getDefault();//create object of
smsmanager

smsmanager.sendTextMessage(number,null,sms,null,nu
ll);

Toast.makeText(getApplicationContext(),"Message
sent success",Toast.LENGTH_SHORT).show();
        }
        catch(Exception e)
        {
            e.printStackTrace();
        }
    }
}
}) ;
}
}
```
